

glasstech, inc.

Service Bulletin

Customer Service Department

TELEPHONE: 419-661-9000 FAX: 419-661-9639

Lubrication of Ramsey Silent Chain

Reference 006

Ramsey silent chain has more sliding friction than other types of chain used on Glasstech tempering furnace systems.

Glasstech has long recommended the use of Alubco (American Lubricants Co.) #20 Nondrip Oil with Moly. This is a petroleum based #20 oil with a nondrip additive and a colloidal particle filler of molybdenum disulfide (MDS_2). The molybdenum disulfide particles "plate out" on the chain wear strip and provide a high lubricity dry film that minimizes the sliding friction of the silent chain.

This dry film extends the life of the wear strips and minimizes the "stick and slip" jerkiness that can occur in a sliding motion so that conveyors using silent chain will run smoothly.

Some Glasstech customers have reported that molybdenum disulfide filled oils are unavailable from their regular lubricant suppliers.

The following discussion will rank some alternative types of lubrication and list pros and cons of each.

Alternates to
Alubco #20
Nondrip Moly

Type of lubricant	Example	Pros	Cons
molybdenum disulfide filled grease	BP-- Energrease Moly EP-2; Mobil Oil-- Mobilgrease Special	similar film strength and film lubricity to Alubco, good slow speed operation due to "slipperiness" of film	requires modification to chain lubrication equipment, difficult to get lubricant into link pin areas
diesel engine oil	Shell Rotella T	contains MdS_2 and graphite additives,	contains less MdS_2 than Alubco, not designed to operate as a dry film lubricant, good film strength and smooth conveyor operation may require excessive oil application which can cause roll slipping
graphite filled oil (synthetic)	Dow Corning C-40 (high temperature chain oil)	excellent temperature resistance, graphite plates out	graphite film strength less than MdS_2 , requires temperatures above 177 degC to evaporate carrier fluid and leave dry film, operation between 121 degC and 177 degC may cause gumming
synthetic oils	Mobil-- SHC626; Mobil 1	good availability	does not have dry lubricating film, may require excessive application rates to maintain smooth conveyor operation which can cause roll slipping